

Biostat 200C Homework 2

Due Apr 25 @ 11:59PM

Kiana Mohammadinik

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To submit homework, please upload both Rmd and html files to Bruinlearn by the deadline.

0.1 Q1. CFR of COVID-19 (90pts)

Of primary interest to public is the risk of dying from COVID-19. A commonly used measure is case fatality rate/ratio/risk (CFR), which is defined as

$$\frac{\text{number of deaths from disease}}{\text{number of diagnosed cases of disease}}.$$

Apparently CFR is not a fixed constant; it changes with time, location, and other factors. Also CFR is different from the infection fatality rate (IFR), the probability that someone infected with COVID-19 dies from it.

In this exercise, we use logistic regression to study how US county-level CFR changes according to demographic information and some health-, education-, and economy-indicators.

0.1.1 Data sources

- 04-04-2020.csv.gz: The data on COVID-19 confirmed cases and deaths on 2020-04-04 is retrieved from the [Johns Hopkins COVID-19 data repository](#). It was downloaded from this [link](#) (commit 0174f38). This repository has been archived by the owner on Mar 10, 2023. It is now read-only. You can download data from box: <https://ucla.box.com/s/brb3vz4nwoq8pjkcutexncymqw583d39l>

- `us-county-health-rankings-2020.csv.gz`: The 2020 County Health Ranking Data was released by [County Health Rankings](#). The data was downloaded from the [Kaggle Uncover COVID-19 Challenge](#) (version 1). You can download data from box: <https://ucla.box.com/s/brb3vz4nwoq8pjkcutexncymqw583d39l>

0.1.2 data preparation

Load the tidyverse package for data manipulation and visualization.

```
# tidyverse of data manipulation and visualization
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.3.1

Warning: package 'tidyr' was built under R version 4.3.1

Warning: package 'purrr' was built under R version 4.3.3

Warning: package 'dplyr' was built under R version 4.3.1

Warning: package 'stringr' was built under R version 4.3.1

Read in the data of COVID-19 cases reported on 2020-04-04.

```
county_count <- read_csv("04-04-2020.csv") %>%
  # cast fips into dbl for use as a key for joining tables
  mutate(FIPS = as.numeric(FIPS)) %>%
  filter(Country_Region == "US") %>%
  print(width = Inf)
```

A tibble: 2,421 x 12

	FIPS	Admin2	Province_State	Country_Region	Last_Update	Lat
	<dbl>	<chr>	<chr>	<chr>	<dtm>	<dbl>
1	45001	Abbeville	South Carolina	US	2020-04-04 23:34:21	34.2
2	22001	Acadia	Louisiana	US	2020-04-04 23:34:21	30.3
3	51001	Accomack	Virginia	US	2020-04-04 23:34:21	37.8
4	16001	Ada	Idaho	US	2020-04-04 23:34:21	43.5
5	19001	Adair	Iowa	US	2020-04-04 23:34:21	41.3
6	21001	Adair	Kentucky	US	2020-04-04 23:34:21	37.1

```

7 29001 Adair      Missouri      US              2020-04-04 23:34:21 40.2
8 40001 Adair      Oklahoma       US              2020-04-04 23:34:21 35.9
9  8001 Adams      Colorado       US              2020-04-04 23:34:21 39.9
10 16003 Adams      Idaho          US              2020-04-04 23:34:21 44.9
    Long_ Confirmed Deaths Recovered Active Combined_Key
    <dbl>      <dbl>   <dbl>      <dbl>   <dbl> <chr>
1  -82.5         6       0         0       0 Abbeville, South Carolina, US
2  -92.4        65       2         0       0 Acadia, Louisiana, US
3  -75.6         8       0         0       0 Accomack, Virginia, US
4 -116.        360       3         0       0 Ada, Idaho, US
5  -94.5         1       0         0       0 Adair, Iowa, US
6  -85.3         3       0         0       0 Adair, Kentucky, US
7  -92.6        10       0         0       0 Adair, Missouri, US
8  -94.7        14       0         0       0 Adair, Oklahoma, US
9 -104.        294       9         0       0 Adams, Colorado, US
10 -116.         1       0         0       0 Adams, Idaho, US
# i 2,411 more rows

```

Standardize the variable names by changing them to lower case.

```
names(county_count) <- str_to_lower(names(county_count))
```

Sanity check by displaying the unique US states and territories:

```

county_count %>%
  select(province_state) %>%
  distinct() %>%
  arrange(province_state) %>%
  print(n = Inf)

```

```

# A tibble: 58 x 1
  province_state
  <chr>
1 Alabama
2 Alaska
3 Arizona
4 Arkansas
5 California
6 Colorado
7 Connecticut
8 Delaware

```

9 Diamond Princess
10 District of Columbia
11 Florida
12 Georgia
13 Grand Princess
14 Guam
15 Hawaii
16 Idaho
17 Illinois
18 Indiana
19 Iowa
20 Kansas
21 Kentucky
22 Louisiana
23 Maine
24 Maryland
25 Massachusetts
26 Michigan
27 Minnesota
28 Mississippi
29 Missouri
30 Montana
31 Nebraska
32 Nevada
33 New Hampshire
34 New Jersey
35 New Mexico
36 New York
37 North Carolina
38 North Dakota
39 Northern Mariana Islands
40 Ohio
41 Oklahoma
42 Oregon
43 Pennsylvania
44 Puerto Rico
45 Recovered
46 Rhode Island
47 South Carolina
48 South Dakota
49 Tennessee
50 Texas
51 Utah

52 Vermont
 53 Virgin Islands
 54 Virginia
 55 Washington
 56 West Virginia
 57 Wisconsin
 58 Wyoming

We want to exclude entries from Diamond Princess, Grand Princess, Guam, Northern Mariana Islands, Puerto Rico, Recovered, and Virgin Islands, and only consider counties from 50 states and DC.

```
county_count <- county_count %>%
  filter(!(province_state %in% c("Diamond Princess", "Grand Princess",
                                "Recovered", "Guam", "Northern Mariana Islands",
                                "Puerto Rico", "Virgin Islands"))) %>%
  print(width = Inf)
```

A tibble: 2,413 x 12

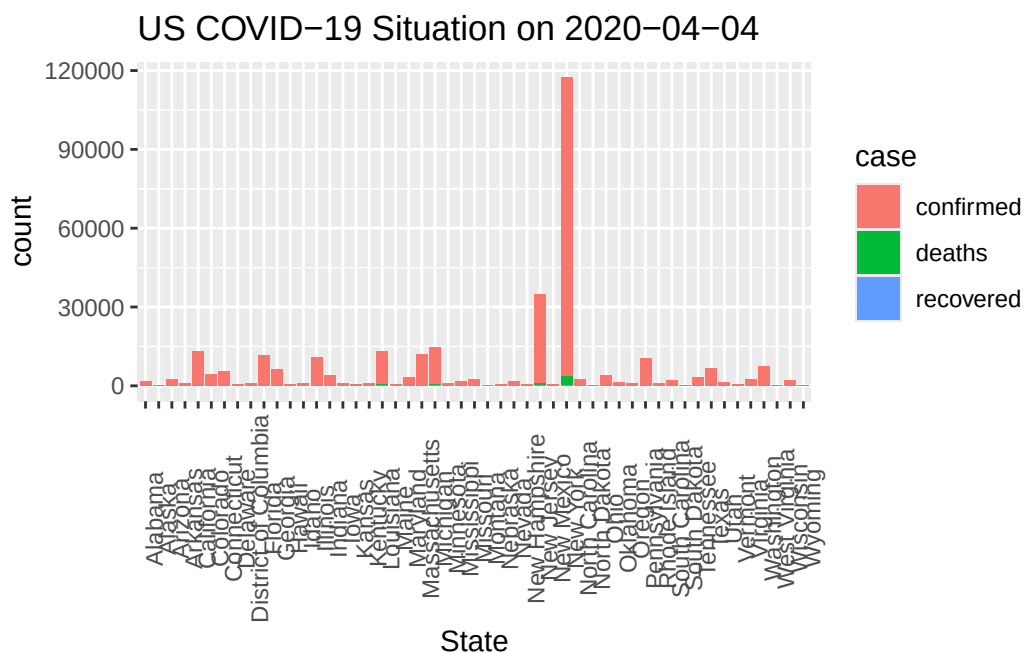
	fips	admin2	province_state	country_region	last_update	lat
	<dbl>	<chr>	<chr>	<chr>	<dtm>	<dbl>
1	45001	Abbeville	South Carolina	US	2020-04-04 23:34:21	34.2
2	22001	Acadia	Louisiana	US	2020-04-04 23:34:21	30.3
3	51001	Accomack	Virginia	US	2020-04-04 23:34:21	37.8
4	16001	Ada	Idaho	US	2020-04-04 23:34:21	43.5
5	19001	Adair	Iowa	US	2020-04-04 23:34:21	41.3
6	21001	Adair	Kentucky	US	2020-04-04 23:34:21	37.1
7	29001	Adair	Missouri	US	2020-04-04 23:34:21	40.2
8	40001	Adair	Oklahoma	US	2020-04-04 23:34:21	35.9
9	8001	Adams	Colorado	US	2020-04-04 23:34:21	39.9
10	16003	Adams	Idaho	US	2020-04-04 23:34:21	44.9

	long_	confirmed	deaths	recovered	active	combined_key
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	-82.5	6	0	0	0	Abbeville, South Carolina, US
2	-92.4	65	2	0	0	Acadia, Louisiana, US
3	-75.6	8	0	0	0	Accomack, Virginia, US
4	-116.	360	3	0	0	Ada, Idaho, US
5	-94.5	1	0	0	0	Adair, Iowa, US
6	-85.3	3	0	0	0	Adair, Kentucky, US
7	-92.6	10	0	0	0	Adair, Missouri, US
8	-94.7	14	0	0	0	Adair, Oklahoma, US
9	-104.	294	9	0	0	Adams, Colorado, US

```
10 -116.      1      0      0      0 Adams, Idaho, US
# i 2,403 more rows
```

Graphical summarize the COVID-19 confirmed cases and deaths on 2020-04-04 by state.

```
county_count %>%
  # turn into long format for easy plotting
  pivot_longer(confirmed:recovered,
               names_to = "case",
               values_to = "count") %>%
  group_by(province_state) %>%
  ggplot() +
  geom_col(mapping = aes(x = province_state, y = `count`, fill = `case`)) +
  # scale_y_log10() +
  labs(title = "US COVID-19 Situation on 2020-04-04", x = "State") +
  theme(axis.text.x = element_text(angle = 90))
```



Read in the 2020 county-level health ranking data.

```
county_info <- read_csv("us-county-health-rankings-2020.csv") %>%
  filter(!is.na(county)) %>%
```

```

# cast fips into dbl for use as a key for joining tables
mutate(fips = as.numeric(fips)) %>%
select(fips,
      state,
      county,
      percent_fair_or_poor_health,
      percent_smokers,
      percent_adults_with_obesity,
      # food_environment_index,
      percent_with_access_to_exercise_opportunities,
      percent_excessive_drinking,
      # teen_birth_rate,
      percent_uninsured,
      # primary_care_physicians_rate,
      # preventable_hospitalization_rate,
      # high_school_graduation_rate,
      percent_some_college,
      percent_unemployed,
      percent_children_in_poverty,
      # `80th_percentile_income`,
      # `20th_percentile_income`,
      percent_single_parent_households,
      # violent_crime_rate,
      percent_severe_housing_problems,
      overcrowding,
      # life_expectancy,
      # age_adjusted_death_rate,
      percent_adults_with_diabetes,
      # hiv_prevalence_rate,
      percent_food_insecure,
      # percent_limited_access_to_healthy_foods,
      percent_insufficient_sleep,
      percent_uninsured_2,
      median_household_income,
      average_traffic_volume_per_meter_of_major_roadways,
      percent_homeowners,
      # percent_severe_housing_cost_burden,
      population_2,
      percent_less_than_18_years_of_age,
      percent_65_and_over,
      percent_black,
      percent_asian,

```



```

    percent_hispanic,
    percent_female,
    percent_rural) %>%
print(width = Inf)

```

A tibble: 3,142 x 30

	fips	state	county	percent_fair_or_poor_health	percent_smokers
	<dbl>	<chr>	<chr>	<dbl>	<dbl>
1	1001	Alabama	Autauga	20.9	18.1
2	1003	Alabama	Baldwin	17.5	17.5
3	1005	Alabama	Barbour	29.6	22.0
4	1007	Alabama	Bibb	19.4	19.1
5	1009	Alabama	Blount	21.7	19.2
6	1011	Alabama	Bullock	31.0	22.9
7	1013	Alabama	Butler	27.9	21.8
8	1015	Alabama	Calhoun	23.1	20.6
9	1017	Alabama	Chambers	24.0	19.4
10	1019	Alabama	Cherokee	20.7	17.5

	percent_adults_with_obesity	percent_with_access_to_exercise_opportunities
	<dbl>	<dbl>
1	33.3	69.1
2	31	73.7
3	41.7	53.2
4	37.6	16.3
5	33.8	15.6
6	37.2	2.50
7	43.3	48.6
8	38.5	47.7
9	40.1	61.9
10	35	33.4

	percent_excessive_drinking	percent_uninsured	percent_some_college
	<dbl>	<dbl>	<dbl>
1	15.0	8.72	62.0
2	18.0	11.3	67.4
3	12.8	12.2	34.9
4	15.6	10.2	44.1
5	14.2	13.4	53.4
6	12.1	11.4	35.0
7	11.9	11.2	41.7
8	13.8	11.9	59.2
9	12.7	11.9	48.5
10	14.1	11.2	51.8

	percent_unemployed	percent_children_in_poverty
	<dbl>	<dbl>
1	3.63	19.3
2	3.62	13.9
3	5.17	43.9
4	3.97	27.8
5	3.51	18
6	4.69	68.3
7	4.79	36.3
8	4.65	26.5
9	3.91	30.7
10	3.57	24.7

	percent_single_parent_households	percent_severe_housing_problems	overcrowding
	<dbl>	<dbl>	<dbl>
1	26.2	14.7	1.20
2	24.1	13.6	1.27
3	56.6	14.6	1.69
4	28.7	10.5	0.255
5	28.6	10.5	1.89
6	74.8	18.1	0.113
7	52.7	13.2	1.69
8	40.2	13.7	1.54
9	46.6	16.0	4.04
10	23.8	13	1.5

	percent_adults_with_diabetes	percent_food_insecure	percent_insufficient_sleep
	<dbl>	<dbl>	<dbl>
1	11.1	13.2	35.9
2	10.7	11.6	33.3
3	17.6	22	38.6
4	14.5	14.3	38.1
5	17	10.7	35.9
6	23.7	24.8	45.0
7	19.2	20.6	41.9
8	17.5	15.7	41.3
9	19.9	17.9	37.3
10	15.2	12.5	35.4

	percent_uninsured_2	median_household_income
	<dbl>	<dbl>
1	11.1	59338
2	14.3	57588
3	16.1	34382
4	13	46064
5	17.1	50412

6	15.2	29267
7	14.5	37365
8	15.4	45400
9	15.2	39917
10	13.9	42132
average_traffic_volume_per_meter_of_major_roadways percent_homeowners		
	<dbl>	<dbl>
1	88.5	74.9
2	87.0	73.6
3	102.	61.4
4	29.3	75.1
5	33.4	78.6
6	4.07	75.5
7	19.3	69.9
8	110.	69.5
9	20.3	67.8
10	25.9	79.0
population_2 percent_less_than_18_years_of_age percent_65_and_over		
	<dbl>	<dbl>
1	55601	23.7
2	218022	21.6
3	24881	20.9
4	22400	20.5
5	57840	23.2
6	10138	21.1
7	19680	22.2
8	114277	21.6
9	33615	20.8
10	26032	19.2
percent_black percent_asian percent_hispanic percent_female percent_rural		
	<dbl>	<dbl>
1	19.3	1.22
2	8.78	1.15
3	48.0	0.454
4	21.1	0.237
5	1.46	0.320
6	69.5	0.187
7	44.6	1.32
8	20.9	0.964
9	39.6	1.33
10	4.24	0.338

i 3,132 more rows

For stability in estimating CFR, we restrict to counties with ≥ 5 confirmed cases.

```
county_count <- county_count %>%
  filter(confirmed >= 5)
```

We join the COVID-19 count data and county-level information using FIPS (Federal Information Processing System) as key.

```
county_data <- county_count %>%
  left_join(county_info, by = "fips") %>%
  print(width = Inf)
```

A tibble: 1,466 x 41

	fips	admin2	province_state	country_region	last_update	lat
	<dbl>	<chr>	<chr>	<chr>	<dtm>	<dbl>
1	45001	Abbeville	South Carolina	US	2020-04-04 23:34:21	34.2
2	22001	Acadia	Louisiana	US	2020-04-04 23:34:21	30.3
3	51001	Accomack	Virginia	US	2020-04-04 23:34:21	37.8
4	16001	Ada	Idaho	US	2020-04-04 23:34:21	43.5
5	29001	Adair	Missouri	US	2020-04-04 23:34:21	40.2
6	40001	Adair	Oklahoma	US	2020-04-04 23:34:21	35.9
7	8001	Adams	Colorado	US	2020-04-04 23:34:21	39.9
8	28001	Adams	Mississippi	US	2020-04-04 23:34:21	31.5
9	31001	Adams	Nebraska	US	2020-04-04 23:34:21	40.5
10	42001	Adams	Pennsylvania	US	2020-04-04 23:34:21	39.9

	long_	confirmed	deaths	recovered	active	combined_key
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	-82.5	6	0	0	0	Abbeville, South Carolina, US
2	-92.4	65	2	0	0	Acadia, Louisiana, US
3	-75.6	8	0	0	0	Accomack, Virginia, US
4	-116.	360	3	0	0	Ada, Idaho, US
5	-92.6	10	0	0	0	Adair, Missouri, US
6	-94.7	14	0	0	0	Adair, Oklahoma, US
7	-104.	294	9	0	0	Adams, Colorado, US
8	-91.4	16	0	0	0	Adams, Mississippi, US
9	-98.5	8	0	0	0	Adams, Nebraska, US
10	-77.2	21	0	0	0	Adams, Pennsylvania, US

	state	county	percent_fair_or_poor_health	percent_smokers
	<chr>	<chr>	<dbl>	<dbl>
1	South Carolina	Abbeville	19.9	17.3
2	Louisiana	Acadia	20.9	21.5
3	Virginia	Accomack	20.1	18.3

4	Idaho	Ada	11.5	12.0
5	Missouri	Adair	21.4	20.5
6	Oklahoma	Adair	28.5	27.7
7	Colorado	Adams	16.6	16.3
8	Mississippi	Adams	27.3	22.2
9	Nebraska	Adams	15.8	14.6
10	Pennsylvania	Adams	15.3	16.2
	percent_adults_with_obesity	percent_with_access_to_exercise_opportunities		
		<dbl>		<dbl>
1		36.7		59.0
2		38.4		42.5
3		36.3		37.4
4		25.6		89.5
5		27.9		78.3
6		47.7		28.5
7		27.8		93.1
8		35.3		69.1
9		36.7		81.6
10		35.6		60.6
	percent_excessive_drinking	percent_uninsured	percent_some_college	
		<dbl>	<dbl>	<dbl>
1		15.9	12.9	52.5
2		19.8	10.7	43.6
3		15.5	16.6	45.1
4		17.9	8.74	73.8
5		18.9	10.6	65.3
6		11.8	24.5	35.1
7		18.9	11.0	57.0
8		12.3	15.0	41.7
9		18.5	8.76	70.8
10		19.2	7.49	57.3
	percent_unemployed	percent_children_in_poverty		
		<dbl>		<dbl>
1		3.98		30.8
2		5.37		35.4
3		3.81		27
4		2.46		10.2
5		3.51		19.9
6		4.17		34.9
7		3.47		12.6
8		6.21		40.4
9		2.87		14.4
10		3.27		11.2

	percent_single_parent_households	percent_severe_housing_problems	overcrowding
	<dbl>	<dbl>	<dbl>
1	37.1	14.3	0.463
2	33.4	12.3	3.51
3	45.9	15.1	2.10
4	23.8	14.0	1.46
5	29.5	18.0	0.740
6	38.3	15.4	5.65
7	31.0	18.1	5.37
8	66.4	12.8	2.37
9	26.2	10.5	0.904
10	26.7	12.3	1.88

	percent_adults_with_diabetes	percent_food_insecure	percent_insufficient_sleep
	<dbl>	<dbl>	<dbl>
1	15.8	15.2	36.1
2	11.4	15.1	32.4
3	15.9	14.1	36.8
4	7.9	12	26.3
5	8.4	17.5	31.9
6	24.3	19.1	39.5
7	7.7	8	31.0
8	13.2	24.7	41.1
9	11	11.7	30.1
10	8.5	8.3	34.7

	percent_uninsured_2	median_household_income
	<dbl>	<dbl>
1	15.9	42412
2	14.0	40484
3	19.4	42879
4	11.1	66827
5	12.3	40395
6	29.6	35156
7	13.8	70199
8	18.7	33392
9	10.7	55167
10	8.46	62877

	average_traffic_volume_per_meter_of_major_roadways	percent_homeowners
	<dbl>	<dbl>
1	11.6	76.3
2	63.7	70.8
3	60.0	67.9
4	277.	68.4
5	45.8	60.0

```

6              16.7              68.6
7              490.              65.2
8              150.              61.7
9              53.4              68.2
10             113.              77.2
  population_2 percent_less_than_18_years_of_age percent_65_and_over
              <dbl>              <dbl>              <dbl>
1           24541              20.1              21.8
2           62190              25.8              15.3
3           32412              20.5              23.6
4          469966              23.8              14.4
5           25339              18.4              14.8
6           22082              26.6              15.9
7          511868              26.5              10.5
8           31192              20.1              18.8
9           31511              23.7              18.2
10          102811              20.0              20.4
  percent_black percent_asian percent_hispanic percent_female percent_rural
              <dbl>          <dbl>          <dbl>          <dbl>          <dbl>
1           27.5           0.412           1.54           51.6           78.6
2           17.9           0.320           2.73           51.2           51.7
3           28.0           0.781           9.34           51.2          100
4            1.24           2.81           8.31           49.9            5.47
5            2.85           2.28           2.57           51.9           37.9
6            0.534          0.802           6.82           50.1           83.3
7            3.19           4.37          40.4           49.5            3.62
8           52.4           0.513           11.3           47.9           37.2
9            0.996           1.33           10.9           50.2           22.5
10           1.60           0.875           7.11           50.8           53.7
# i 1,456 more rows

```

Numerical summaries of each variable:

```
summary(county_data)
```

fips	admin2	province_state	country_region
Min. : 1001	Length:1466	Length:1466	Length:1466
1st Qu.:18003	Class :character	Class :character	Class :character
Median :29029	Mode :character	Mode :character	Mode :character
Mean :30076			
3rd Qu.:42077			
Max. :90053			

```

NA's      :13
last_update      lat      long_
Min.   :2020-04-04 23:34:21  Min.   :19.60  Min.   : -159.60
1st Qu.:2020-04-04 23:34:21  1st Qu.:33.96  1st Qu.: -94.56
Median :2020-04-04 23:34:21  Median :38.02  Median : -86.48
Mean   :2020-04-04 23:34:21  Mean   :37.71  Mean   : -89.73
3rd Qu.:2020-04-04 23:34:21  3rd Qu.:41.38  3rd Qu.: -81.22
Max.   :2020-04-04 23:34:21  Max.   :64.81  Max.   : -68.65
NA's      :19      NA's      :19

```

```

confirmed      deaths      recovered      active
Min.   :      5.0  Min.   :      0.000  Min.   :0  Min.   :0
1st Qu.:      9.0  1st Qu.:      0.000  1st Qu.:0  1st Qu.:0
Median :     20.0  Median :      0.000  Median :0  Median :0
Mean   :    208.8  Mean   :      4.842  Mean   :0  Mean   :0
3rd Qu.:     68.0  3rd Qu.:      2.000  3rd Qu.:0  3rd Qu.:0
Max.   :   63306.0  Max.   :   1905.000  Max.   :0  Max.   :0

```

```

combined_key      state      county
Length:1466      Length:1466      Length:1466
Class :character  Class :character  Class :character
Mode  :character  Mode  :character  Mode  :character

```

```

percent_fair_or_poor_health percent_smokers percent_adults_with_obesity
Min.   : 8.121      Min.   : 5.909  Min.   :12.40
1st Qu.:14.390      1st Qu.:14.899  1st Qu.:29.10
Median :17.010      Median :17.147  Median :32.95
Mean   :17.594      Mean   :17.153  Mean   :32.41
3rd Qu.:20.377      3rd Qu.:19.365  3rd Qu.:36.20
Max.   :38.887      Max.   :27.775  Max.   :51.00
NA's   :28          NA's   :28      NA's   :28

```

```

percent_with_access_to_exercise_opportunities percent_excessive_drinking
Min.   : 0.00      Min.   : 7.81
1st Qu.: 59.95      1st Qu.:15.70
Median : 74.71      Median :18.03
Mean   : 71.14      Mean   :17.92
3rd Qu.: 85.94      3rd Qu.:20.00
Max.   :100.00      Max.   :28.62
NA's   :28          NA's   :28

```

```

percent_uninsured percent_some_college percent_unemployed
Min.   : 2.263      Min.   :30.06      Min.   : 1.582

```


1st Qu.: 6.754	1st Qu.:53.24	1st Qu.: 3.252
Median : 9.925	Median :61.21	Median : 3.870
Mean :10.583	Mean :60.87	Mean : 4.071
3rd Qu.:13.519	3rd Qu.:68.74	3rd Qu.: 4.690
Max. :31.208	Max. :90.34	Max. :18.092
NA's :28	NA's :28	NA's :28

percent_children_in_poverty	percent_single_parent_households
Min. : 2.50	Min. : 9.43
1st Qu.:12.82	1st Qu.:27.09
Median :18.40	Median :32.96
Mean :19.46	Mean :33.84
3rd Qu.:24.50	3rd Qu.:38.94
Max. :55.00	Max. :80.00
NA's :28	NA's :28

percent_severe_housing_problems	overcrowding	percent_adults_with_diabetes
Min. : 6.562	Min. : 0.000	Min. : 1.800
1st Qu.:12.267	1st Qu.: 1.378	1st Qu.: 9.125
Median :14.439	Median : 1.962	Median :11.300
Mean :15.079	Mean : 2.429	Mean :11.749
3rd Qu.:16.976	3rd Qu.: 2.882	3rd Qu.:13.900
Max. :33.391	Max. :14.489	Max. :34.100
NA's :28	NA's :28	NA's :28

percent_food_insecure	percent_insufficient_sleep	percent_uninsured_2
Min. : 3.40	Min. :23.03	Min. : 2.683
1st Qu.:10.70	1st Qu.:31.42	1st Qu.: 7.865
Median :12.70	Median :34.02	Median :12.027
Mean :13.26	Mean :33.88	Mean :12.776
3rd Qu.:15.20	3rd Qu.:36.56	3rd Qu.:16.541
Max. :33.50	Max. :46.71	Max. :42.397
NA's :28	NA's :28	NA's :28

median_household_income	average_traffic_volume_per_meter_of_major_roadways
Min. : 25385	Min. : 0.00
1st Qu.: 46994	1st Qu.: 53.05
Median : 54317	Median : 105.00
Mean : 57584	Mean : 201.39
3rd Qu.: 64754	3rd Qu.: 206.92
Max. :140382	Max. :4444.12
NA's :28	NA's :28

percent_homeowners	population_2	percent_less_than_18_years_of_age
Min. :24.13	Min. : 2887	Min. : 7.069
1st Qu.:64.34	1st Qu.: 36502	1st Qu.:20.321
Median :69.98	Median : 75478	Median :22.182
Mean :68.98	Mean : 202450	Mean :22.197

3rd Qu.:74.78	3rd Qu.: 180031	3rd Qu.:24.002	
Max. :89.76	Max. :10105518	Max. :35.447	
NA's :28	NA's :28	NA's :28	
percent_65_and_over	percent_black	percent_asian	percent_hispanic
Min. : 7.722	Min. : 0.1286	Min. : 0.06245	Min. : 0.7952
1st Qu.:14.927	1st Qu.: 1.6175	1st Qu.: 0.68248	1st Qu.: 2.9419
Median :17.222	Median : 5.6397	Median : 1.23421	Median : 5.5939
Mean :17.516	Mean :12.4178	Mean : 2.40412	Mean :10.0010
3rd Qu.:19.598	3rd Qu.:17.5931	3rd Qu.: 2.67550	3rd Qu.:11.0564
Max. :57.587	Max. :81.9544	Max. :42.95231	Max. :96.3595
NA's :28	NA's :28	NA's :28	NA's :28
percent_female	percent_rural		
Min. :34.63	Min. : 0.00		
1st Qu.:50.00	1st Qu.: 17.11		
Median :50.66	Median : 36.97		
Mean :50.46	Mean : 40.11		
3rd Qu.:51.35	3rd Qu.: 60.00		
Max. :56.87	Max. :100.00		
NA's :28	NA's :28		

List rows in county_data that don't have a match in county_count:

```
county_data %>%
  filter(is.na(state) & is.na(county)) %>%
  print(n = Inf)
```

A tibble: 28 x 41

	fips	admin2	province_state	country_region	last_update	lat	long_
	<dbl>	<chr>	<chr>	<chr>	<dtm>	<dbl>	<dbl>
1	NA	DeKalb	Tennessee	US	2020-04-04 23:34:21	36.0	-85.8
2	NA	DeSoto	Florida	US	2020-04-04 23:34:21	27.2	-81.8
3	NA	Dukes a~	Massachusetts	US	2020-04-04 23:34:21	41.4	-70.7
4	NA	Fillmore	Minnesota	US	2020-04-04 23:34:21	43.7	-92.1
5	NA	Kansas ~	Missouri	US	2020-04-04 23:34:21	39.1	-94.6
6	NA	LaSalle	Illinois	US	2020-04-04 23:34:21	41.3	-88.9
7	NA	Manassas	Virginia	US	2020-04-04 23:34:21	38.7	-77.5
8	NA	McDuffie	Georgia	US	2020-04-04 23:34:21	33.5	-82.5
9	NA	Out of ~	Michigan	US	2020-04-04 23:34:21	NA	NA
10	NA	Out of ~	Tennessee	US	2020-04-04 23:34:21	NA	NA
11	90005	Unassig~	Arkansas	US	2020-04-04 23:34:21	NA	NA
12	90008	Unassig~	Colorado	US	2020-04-04 23:34:21	NA	NA
13	90009	Unassig~	Connecticut	US	2020-04-04 23:34:21	NA	NA

```

14 90013 Unassig~ Georgia      US      2020-04-04 23:34:21 NA      NA
15 90015 Unassig~ Hawaii      US      2020-04-04 23:34:21 NA      NA
16 90017 Unassig~ Illinois    US      2020-04-04 23:34:21 NA      NA
17 90021 Unassig~ Kentucky    US      2020-04-04 23:34:21 NA      NA
18 NA Unassig~ Louisiana      US      2020-04-04 23:34:21 NA      NA
19 90023 Unassig~ Maine        US      2020-04-04 23:34:21 NA      NA
20 90025 Unassig~ Massachusetts US      2020-04-04 23:34:21 NA      NA
21 NA Unassig~ Michigan       US      2020-04-04 23:34:21 NA      NA
22 90032 Unassig~ Nevada       US      2020-04-04 23:34:21 NA      NA
23 90034 Unassig~ New Jersey   US      2020-04-04 23:34:21 NA      NA
24 90044 Unassig~ Rhode Island US      2020-04-04 23:34:21 NA      NA
25 90047 Unassig~ Tennessee   US      2020-04-04 23:34:21 NA      NA
26 90050 Unassig~ Vermont      US      2020-04-04 23:34:21 NA      NA
27 90053 Unassig~ Washington   US      2020-04-04 23:34:21 NA      NA
28 NA Weber Utah              US      2020-04-04 23:34:21 41.3 -112.
# i 34 more variables: confirmed <dbl>, deaths <dbl>, recovered <dbl>,
# active <dbl>, combined_key <chr>, state <chr>, county <chr>,
# percent_fair_or_poor_health <dbl>, percent_smokers <dbl>,
# percent_adults_with_obesity <dbl>,
# percent_with_access_to_exercise_opportunities <dbl>,
# percent_excessive_drinking <dbl>, percent_uninsured <dbl>,
# percent_some_college <dbl>, percent_unemployed <dbl>, ...

```

We found there are some rows that miss fips.

```

county_count %>%
  filter(is.na(fips)) %>%
  select(fips, admin2, province_state) %>%
  print(n = Inf)

```

```

# A tibble: 13 x 3
   fips admin2      province_state
  <dbl> <chr>      <chr>
1 NA DeKalb      Tennessee
2 NA DeSoto      Florida
3 NA Dukes and Nantucket Massachusetts
4 NA Fillmore    Minnesota
5 NA Kansas City Missouri
6 NA LaSalle     Illinois
7 NA Manassas    Virginia
8 NA McDuffie    Georgia
9 NA Out of MI   Michigan

```

```

10    NA Out of TN           Tennessee
11    NA Unassigned         Louisiana
12    NA Unassigned         Michigan
13    NA Weber              Utah

```

We need to (1) manually set the `fips` for some counties, (2) discard those `Unassigned`, `unassigned` or `Out of`, and (3) try to join with `county_info` again.

```

county_data <- county_count %>%
  # manually set FIPS for some counties
  mutate(fips = ifelse(admin2 == "DeKalb" & province_state == "Tennessee", 47041, fips)) %>%
  mutate(fips = ifelse(admin2 == "DeSoto" & province_state == "Florida", 12027, fips)) %>%
  #mutate(fips = ifelse(admin2 == "Dona Ana" & province_state == "New Mexico", 35013, fips))
  mutate(fips = ifelse(admin2 == "Dukes and Nantucket" & province_state == "Massachusetts", 25013, fips)) %>%
  mutate(fips = ifelse(admin2 == "Fillmore" & province_state == "Minnesota", 27045, fips)) %>%
  #mutate(fips = ifelse(admin2 == "Harris" & province_state == "Texas", 48201, fips)) %>%
  #mutate(fips = ifelse(admin2 == "Kenai Peninsula" & province_state == "Alaska", 2122, fips)) %>%
  mutate(fips = ifelse(admin2 == "LaSalle" & province_state == "Illinois", 17099, fips)) %>%
  #mutate(fips = ifelse(admin2 == "LaSalle" & province_state == "Louisiana", 22059, fips)) %>%
  #mutate(fips = ifelse(admin2 == "Lac qui Parle" & province_state == "Minnesota", 27073, fips)) %>%
  mutate(fips = ifelse(admin2 == "Manassas" & province_state == "Virginia", 51683, fips)) %>%
  #mutate(fips = ifelse(admin2 == "Matanuska-Susitna" & province_state == "Alaska", 2170, fips)) %>%
  mutate(fips = ifelse(admin2 == "McDuffie" & province_state == "Georgia", 13189, fips)) %>%
  #mutate(fips = ifelse(admin2 == "McIntosh" & province_state == "Georgia", 13191, fips)) %>%
  #mutate(fips = ifelse(admin2 == "McKean" & province_state == "Pennsylvania", 42083, fips)) %>%
  mutate(fips = ifelse(admin2 == "Weber" & province_state == "Utah", 49057, fips)) %>%
  filter(!(is.na(fips) | str_detect(admin2, "Out of") | str_detect(admin2, "Unassigned"))) %>%
  left_join(county_info, by = "fips") %>%
  print(width = Inf)

```

A tibble: 1,446 x 41

	fips	admin2	province_state	country_region	last_update	lat
	<dbl>	<chr>	<chr>	<chr>	<dtm>	<dbl>
1	45001	Abbeville	South Carolina	US	2020-04-04 23:34:21	34.2
2	22001	Acadia	Louisiana	US	2020-04-04 23:34:21	30.3
3	51001	Accomack	Virginia	US	2020-04-04 23:34:21	37.8
4	16001	Ada	Idaho	US	2020-04-04 23:34:21	43.5
5	29001	Adair	Missouri	US	2020-04-04 23:34:21	40.2
6	40001	Adair	Oklahoma	US	2020-04-04 23:34:21	35.9
7	8001	Adams	Colorado	US	2020-04-04 23:34:21	39.9
8	28001	Adams	Mississippi	US	2020-04-04 23:34:21	31.5
9	31001	Adams	Nebraska	US	2020-04-04 23:34:21	40.5

```

10 42001 Adams      Pennsylvania  US      2020-04-04 23:34:21  39.9
    long_ confirmed deaths recovered active combined_key
    <dbl>      <dbl>  <dbl>      <dbl>  <dbl> <chr>
1  -82.5        6      0          0      0 Abbeville, South Carolina, US
2  -92.4       65      2          0      0 Acadia, Louisiana, US
3  -75.6        8      0          0      0 Accomack, Virginia, US
4 -116.       360      3          0      0 Ada, Idaho, US
5  -92.6       10      0          0      0 Adair, Missouri, US
6  -94.7       14      0          0      0 Adair, Oklahoma, US
7 -104.       294      9          0      0 Adams, Colorado, US
8  -91.4       16      0          0      0 Adams, Mississippi, US
9  -98.5        8      0          0      0 Adams, Nebraska, US
10 -77.2       21      0          0      0 Adams, Pennsylvania, US
    state      county  percent_fair_or_poor_health percent_smokers
    <chr>      <chr>      <dbl>      <dbl>
1 South Carolina Abbeville      19.9      17.3
2 Louisiana      Acadia      20.9      21.5
3 Virginia      Accomack      20.1      18.3
4 Idaho          Ada      11.5      12.0
5 Missouri      Adair      21.4      20.5
6 Oklahoma      Adair      28.5      27.7
7 Colorado      Adams      16.6      16.3
8 Mississippi    Adams      27.3      22.2
9 Nebraska      Adams      15.8      14.6
10 Pennsylvania  Adams      15.3      16.2
    percent_adults_with_obesity percent_with_access_to_exercise_opportunities
    <dbl>      <dbl>
1      36.7      59.0
2      38.4      42.5
3      36.3      37.4
4      25.6      89.5
5      27.9      78.3
6      47.7      28.5
7      27.8      93.1
8      35.3      69.1
9      36.7      81.6
10     35.6      60.6
    percent_excessive_drinking percent_uninsured percent_some_college
    <dbl>      <dbl>      <dbl>
1      15.9      12.9      52.5
2      19.8      10.7      43.6
3      15.5      16.6      45.1
4      17.9      8.74      73.8

```

5	18.9	10.6	65.3
6	11.8	24.5	35.1
7	18.9	11.0	57.0
8	12.3	15.0	41.7
9	18.5	8.76	70.8
10	19.2	7.49	57.3
	percent_unemployed	percent_children_in_poverty	
	<dbl>	<dbl>	
1	3.98	30.8	
2	5.37	35.4	
3	3.81	27	
4	2.46	10.2	
5	3.51	19.9	
6	4.17	34.9	
7	3.47	12.6	
8	6.21	40.4	
9	2.87	14.4	
10	3.27	11.2	
	percent_single_parent_households	percent_severe_housing_problems	overcrowding
	<dbl>	<dbl>	<dbl>
1	37.1	14.3	0.463
2	33.4	12.3	3.51
3	45.9	15.1	2.10
4	23.8	14.0	1.46
5	29.5	18.0	0.740
6	38.3	15.4	5.65
7	31.0	18.1	5.37
8	66.4	12.8	2.37
9	26.2	10.5	0.904
10	26.7	12.3	1.88
	percent_adults_with_diabetes	percent_food_insecure	percent_insufficient_sleep
	<dbl>	<dbl>	<dbl>
1	15.8	15.2	36.1
2	11.4	15.1	32.4
3	15.9	14.1	36.8
4	7.9	12	26.3
5	8.4	17.5	31.9
6	24.3	19.1	39.5
7	7.7	8	31.0
8	13.2	24.7	41.1
9	11	11.7	30.1
10	8.5	8.3	34.7
	percent_uninsured_2	median_household_income	

	<dbl>	<dbl>
1	15.9	42412
2	14.0	40484
3	19.4	42879
4	11.1	66827
5	12.3	40395
6	29.6	35156
7	13.8	70199
8	18.7	33392
9	10.7	55167
10	8.46	62877

	average_traffic_volume_per_meter_of_major_roadways	percent_homeowners
	<dbl>	<dbl>
1	11.6	76.3
2	63.7	70.8
3	60.0	67.9
4	277.	68.4
5	45.8	60.0
6	16.7	68.6
7	490.	65.2
8	150.	61.7
9	53.4	68.2
10	113.	77.2

	population_2	percent_less_than_18_years_of_age	percent_65_and_over
	<dbl>	<dbl>	<dbl>
1	24541	20.1	21.8
2	62190	25.8	15.3
3	32412	20.5	23.6
4	469966	23.8	14.4
5	25339	18.4	14.8
6	22082	26.6	15.9
7	511868	26.5	10.5
8	31192	20.1	18.8
9	31511	23.7	18.2
10	102811	20.0	20.4

	percent_black	percent_asian	percent_hispanic	percent_female	percent_rural
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	27.5	0.412	1.54	51.6	78.6
2	17.9	0.320	2.73	51.2	51.7
3	28.0	0.781	9.34	51.2	100
4	1.24	2.81	8.31	49.9	5.47
5	2.85	2.28	2.57	51.9	37.9
6	0.534	0.802	6.82	50.1	83.3

7	3.19	4.37	40.4	49.5	3.62
8	52.4	0.513	11.3	47.9	37.2
9	0.996	1.33	10.9	50.2	22.5
10	1.60	0.875	7.11	50.8	53.7

i 1,436 more rows

Summarize again

```
summary(county_data)
```

fips	admin2	province_state	country_region
Min. : 1001	Length:1446	Length:1446	Length:1446
1st Qu.:17186	Class :character	Class :character	Class :character
Median :28156	Mode :character	Mode :character	Mode :character
Mean :29455			
3rd Qu.:42048			
Max. :56039			
last_update	lat	long_	
Min. :2020-04-04 23:34:21	Min. :19.60	Min. :-159.60	
1st Qu.:2020-04-04 23:34:21	1st Qu.:33.96	1st Qu.: -94.52	
Median :2020-04-04 23:34:21	Median :38.02	Median : -86.48	
Mean :2020-04-04 23:34:21	Mean :37.71	Mean : -89.73	
3rd Qu.:2020-04-04 23:34:21	3rd Qu.:41.39	3rd Qu.: -81.21	
Max. :2020-04-04 23:34:21	Max. :64.81	Max. : -68.65	
confirmed	deaths	recovered	active
Min. : 5.0	Min. : 0.000	Min. :0	Min. :0
1st Qu.: 9.0	1st Qu.: 0.000	1st Qu.:0	1st Qu.:0
Median : 20.0	Median : 0.000	Median :0	Median :0
Mean : 207.2	Mean : 4.854	Mean :0	Mean :0
3rd Qu.: 66.0	3rd Qu.: 2.000	3rd Qu.:0	3rd Qu.:0
Max. :63306.0	Max. :1905.000	Max. :0	Max. :0
combined_key	state	county	
Length:1446	Length:1446	Length:1446	
Class :character	Class :character	Class :character	
Mode :character	Mode :character	Mode :character	
percent_fair_or_poor_health	percent_smokers	percent_adults_with_obesity	
Min. : 8.121	Min. : 5.909	Min. :12.40	
1st Qu.:14.390	1st Qu.:14.899	1st Qu.:29.10	
Median :17.010	Median :17.143	Median :32.90	

Mean :17.594	Mean :17.151	Mean :32.39
3rd Qu.:20.398	3rd Qu.:19.365	3rd Qu.:36.20
Max. :38.887	Max. :27.775	Max. :51.00
percent_with_access_to_exercise_opportunities	percent_excessive_drinking	
Min. : 0.00	Min. : 7.81	
1st Qu.: 59.95	1st Qu.:15.68	
Median : 74.71	Median :18.03	
Mean : 71.15	Mean :17.91	
3rd Qu.: 85.97	3rd Qu.:20.01	
Max. :100.00	Max. :28.62	
percent_uninsured	percent_some_college	percent_unemployed
Min. : 2.263	Min. :21.14	Min. : 1.582
1st Qu.: 6.754	1st Qu.:53.21	1st Qu.: 3.252
Median : 9.937	Median :61.19	Median : 3.870
Mean :10.592	Mean :60.83	Mean : 4.071
3rd Qu.:13.527	3rd Qu.:68.72	3rd Qu.: 4.690
Max. :31.208	Max. :90.34	Max. :18.092
percent_children_in_poverty	percent_single_parent_households	
Min. : 2.50	Min. : 9.43	
1st Qu.:12.82	1st Qu.:27.07	
Median :18.40	Median :32.96	
Mean :19.46	Mean :33.83	
3rd Qu.:24.50	3rd Qu.:38.93	
Max. :55.00	Max. :80.00	
percent_severe_housing_problems	overcrowding	percent_adults_with_diabetes
Min. : 6.562	Min. : 0.000	Min. : 1.80
1st Qu.:12.267	1st Qu.: 1.379	1st Qu.: 9.10
Median :14.439	Median : 1.971	Median :11.30
Mean :15.082	Mean : 2.437	Mean :11.75
3rd Qu.:16.992	3rd Qu.: 2.887	3rd Qu.:13.90
Max. :33.391	Max. :14.489	Max. :34.10
percent_food_insecure	percent_insufficient_sleep	percent_uninsured_2
Min. : 3.40	Min. :23.03	Min. : 2.683
1st Qu.:10.70	1st Qu.:31.42	1st Qu.: 7.865
Median :12.70	Median :34.02	Median :12.027
Mean :13.25	Mean :33.88	Mean :12.786
3rd Qu.:15.20	3rd Qu.:36.54	3rd Qu.:16.572
Max. :33.50	Max. :46.71	Max. :42.397
median_household_income	average_traffic_volume_per_meter_of_major_roadways	
Min. : 25385	Min. : 0.00	
1st Qu.: 46994	1st Qu.: 53.09	
Median : 54317	Median : 104.63	
Mean : 57600	Mean : 200.72	

3rd Qu.: 64775	3rd Qu.: 206.78		
Max. :140382	Max. :4444.12		
percent_homeowners	population_2	percent_less_than_18_years_of_age	
Min. :24.13	Min. : 2887	Min. : 7.069	
1st Qu.:64.36	1st Qu.: 36275	1st Qu.:20.326	
Median :69.96	Median : 75382	Median :22.182	
Mean :68.99	Mean : 201689	Mean :22.204	
3rd Qu.:74.77	3rd Qu.: 179982	3rd Qu.:24.019	
Max. :89.76	Max. :10105518	Max. :35.447	
percent_65_and_over	percent_black	percent_asian	percent_hispanic
Min. : 7.722	Min. : 0.1286	Min. : 0.06245	Min. : 0.7952
1st Qu.:14.913	1st Qu.: 1.6168	1st Qu.: 0.68228	1st Qu.: 2.9451
Median :17.225	Median : 5.6397	Median : 1.22863	Median : 5.6100
Mean :17.512	Mean :12.4056	Mean : 2.40009	Mean :10.0338
3rd Qu.:19.598	3rd Qu.:17.4904	3rd Qu.: 2.66813	3rd Qu.:11.1199
Max. :57.587	Max. :81.9544	Max. :42.95231	Max. :96.3595
percent_female	percent_rural		
Min. :34.63	Min. : 0.00		
1st Qu.:50.00	1st Qu.: 17.11		
Median :50.65	Median : 36.97		
Mean :50.46	Mean : 40.12		
3rd Qu.:51.35	3rd Qu.: 60.04		
Max. :56.87	Max. :100.00		

If there are variables with missing value for many counties, we go back and remove those variables from consideration.

Let's create a final data frame for analysis.

```
county_data <- county_data %>%
  mutate(state = as.factor(state)) %>%
  select(county, confirmed, deaths, state, percent_fair_or_poor_health:percent_rural)
summary(county_data)
```

county	confirmed	deaths	state
Length:1446	Min. : 5.0	Min. : 0.000	Georgia : 96
Class :character	1st Qu.: 9.0	1st Qu.: 0.000	Texas : 80
Mode :character	Median : 20.0	Median : 0.000	North Carolina: 63
	Mean : 207.2	Mean : 4.854	Mississippi : 61
	3rd Qu.: 66.0	3rd Qu.: 2.000	Indiana : 58
	Max. :63306.0	Max. :1905.000	Ohio : 57
			(Other) :1031

percent_fair_or_poor_health	percent_smokers	percent_adults_with_obesity
Min. : 8.121	Min. : 5.909	Min. :12.40
1st Qu.:14.390	1st Qu.:14.899	1st Qu.:29.10
Median :17.010	Median :17.143	Median :32.90
Mean :17.594	Mean :17.151	Mean :32.39
3rd Qu.:20.398	3rd Qu.:19.365	3rd Qu.:36.20
Max. :38.887	Max. :27.775	Max. :51.00

percent_with_access_to_exercise_opportunities	percent_excessive_drinking
Min. : 0.00	Min. : 7.81
1st Qu.: 59.95	1st Qu.:15.68
Median : 74.71	Median :18.03
Mean : 71.15	Mean :17.91
3rd Qu.: 85.97	3rd Qu.:20.01
Max. :100.00	Max. :28.62

percent_uninsured	percent_some_college	percent_unemployed
Min. : 2.263	Min. :21.14	Min. : 1.582
1st Qu.: 6.754	1st Qu.:53.21	1st Qu.: 3.252
Median : 9.937	Median :61.19	Median : 3.870
Mean :10.592	Mean :60.83	Mean : 4.071
3rd Qu.:13.527	3rd Qu.:68.72	3rd Qu.: 4.690
Max. :31.208	Max. :90.34	Max. :18.092

percent_children_in_poverty	percent_single_parent_households
Min. : 2.50	Min. : 9.43
1st Qu.:12.82	1st Qu.:27.07
Median :18.40	Median :32.96
Mean :19.46	Mean :33.83
3rd Qu.:24.50	3rd Qu.:38.93
Max. :55.00	Max. :80.00

percent_severe_housing_problems	overcrowding	percent_adults_with_diabetes
Min. : 6.562	Min. : 0.000	Min. : 1.80
1st Qu.:12.267	1st Qu.: 1.379	1st Qu.: 9.10
Median :14.439	Median : 1.971	Median :11.30
Mean :15.082	Mean : 2.437	Mean :11.75
3rd Qu.:16.992	3rd Qu.: 2.887	3rd Qu.:13.90
Max. :33.391	Max. :14.489	Max. :34.10

percent_food_insecure	percent_insufficient_sleep	percent_uninsured_2
Min. : 3.40	Min. :23.03	Min. : 2.683
1st Qu.:10.70	1st Qu.:31.42	1st Qu.: 7.865

Median :12.70	Median :34.02	Median :12.027
Mean :13.25	Mean :33.88	Mean :12.786
3rd Qu.:15.20	3rd Qu.:36.54	3rd Qu.:16.572
Max. :33.50	Max. :46.71	Max. :42.397

median_household_income	average_traffic_volume_per_meter_of_major_roadways
Min. : 25385	Min. : 0.00
1st Qu.: 46994	1st Qu.: 53.09
Median : 54317	Median : 104.63
Mean : 57600	Mean : 200.72
3rd Qu.: 64775	3rd Qu.: 206.78
Max. :140382	Max. :4444.12

percent_homeowners	population_2	percent_less_than_18_years_of_age
Min. :24.13	Min. : 2887	Min. : 7.069
1st Qu.:64.36	1st Qu.: 36275	1st Qu.:20.326
Median :69.96	Median : 75382	Median :22.182
Mean :68.99	Mean : 201689	Mean :22.204
3rd Qu.:74.77	3rd Qu.: 179982	3rd Qu.:24.019
Max. :89.76	Max. :10105518	Max. :35.447

percent_65_and_over	percent_black	percent_asian	percent_hispanic
Min. : 7.722	Min. : 0.1286	Min. : 0.06245	Min. : 0.7952
1st Qu.:14.913	1st Qu.: 1.6168	1st Qu.: 0.68228	1st Qu.: 2.9451
Median :17.225	Median : 5.6397	Median : 1.22863	Median : 5.6100
Mean :17.512	Mean :12.4056	Mean : 2.40009	Mean :10.0338
3rd Qu.:19.598	3rd Qu.:17.4904	3rd Qu.: 2.66813	3rd Qu.:11.1199
Max. :57.587	Max. :81.9544	Max. :42.95231	Max. :96.3595

percent_female	percent_rural
Min. :34.63	Min. : 0.00
1st Qu.:50.00	1st Qu.: 17.11
Median :50.65	Median : 36.97
Mean :50.46	Mean : 40.12
3rd Qu.:51.35	3rd Qu.: 60.04
Max. :56.87	Max. :100.00

Display the 10 counties with highest CFR.

```
county_data %>%
  mutate(cfr = deaths / confirmed) %>%
```

```
select(county, state, confirmed, deaths, cfr) %>%
  arrange(desc(cfr)) %>%
  top_n(10)
```

A tibble: 18 x 5

	county	state	confirmed	deaths	cfr
	<chr>	<fct>	<dbl>	<dbl>	<dbl>
1	Emmet	Michigan	7	2	0.286
2	Grand Traverse	Michigan	12	3	0.25
3	Toole	Montana	12	3	0.25
4	Fayette	Indiana	14	3	0.214
5	Concordia	Louisiana	5	1	0.2
6	Harrison	Texas	5	1	0.2
7	Huntington	Indiana	5	1	0.2
8	Isabella	Michigan	10	2	0.2
9	McDuffie	Georgia	5	1	0.2
10	Navarro	Texas	5	1	0.2
11	Orange	Indiana	5	1	0.2
12	Perry	Pennsylvania	5	1	0.2
13	Randolph	Indiana	5	1	0.2
14	Rockingham	North Carolina	5	1	0.2
15	Seneca	Ohio	5	1	0.2
16	Toombs	Georgia	5	1	0.2
17	Vigo	Indiana	10	2	0.2
18	Washington	Alabama	5	1	0.2

Write final data into a csv file for future use.

```
write_csv(county_data, "covid19-county-data-20200404.csv")
```

0.1.3 Note:

Given that the datasets were collected in the middle of the pandemic, what assumptions of CFR might be violated by defining CFR as `deaths/confirmed` from this data set?

Because COVID-19 pandemic was still ongoing in 2020, we should realize some critical assumptions for defining CFR are not met using this datasets.

1. Numbers of confirmed cases do not reflect the number of diagnosed people. This is mainly limited by the availability of testing.
2. Some confirmed cases may die later.

With acknowledgement of these severe limitations, we continue to use `deaths/confirmed` as a very rough proxy of CFR.

0.1.4 Q1.1 (5pts)

Read and run above code to generate a data frame `county_data` that includes county-level COVID-19 confirmed cases and deaths, demographic, and health related information.

0.1.5 Q1.2(5pts)

What assumptions of logistic regression may be violated by this data set? Logistic regression assumes that the log-odds of the outcome is linearly related to the predictors, that observations are independent, and that there is no multicollinearity among predictors, but our dataset may violate some of these assumptions in several ways. First, observations (counties) may not be fully independent due to the shared state/federal-level policies or geographic proximity. Second, some predictors might have nonlinear relationships with the outcome, violating the linearity of the logit. Third, strong correlations among socioeconomic and demographic variables may lead to multicollinearity. Also, counties with very small populations or extreme CFRs may have extra undue influence on the model and lead to high-leverage or outlier effects.

0.1.6 Q1.3 (10pts)

Run a logistic regression, using variables `state`, ..., `percent_rural` as predictors. **Solution:**

```
library(broom)
```

Warning: package 'broom' was built under R version 4.3.3

```
# Logistic regression binomial
full_model <- glm(cbind(deaths, confirmed - deaths) ~ state + percent_fair_or_poor_health +
```

0.1.7 Q1.4 (10pts)

Interpret the regression coefficients of 3 significant predictors with p-value <0.05. **Solution:**

```
tidy(full_model) %>%
  arrange(p.value) %>%
  filter(p.value < 0.05)
```

```
# A tibble: 33 x 5
```

	term <chr>	estimate <dbl>	std.error <dbl>	statistic <dbl>	p.value <dbl>
1	percent_homeowners	-0.0340	0.00615	-5.53	0.0000000322
2	stateVirginia	-1.53	0.318	-4.82	0.00000142
3	percent_insufficient_sleep	0.0444	0.00995	4.46	0.00000812
4	percent_severe_housing_problems	-0.0447	0.0108	-4.16	0.0000321
5	percent_children_in_poverty	0.0303	0.00766	3.95	0.0000775
6	stateNew York	-0.925	0.247	-3.75	0.000176
7	statePennsylvania	-0.807	0.230	-3.51	0.000450
8	stateRhode Island	-2.16	0.626	-3.45	0.000564
9	stateMassachusetts	-0.903	0.265	-3.41	0.000654
10	median_household_income	-0.0000108	0.00000332	-3.27	0.00109

```
# i 23 more rows
```

percent_hispanic: For each 1 percentage point increase in the percent of Hispanic residents in a county, the log-odds of COVID-19 death among confirmed cases decreases by 0.0192, holding other variables constant.

Converting to odds ratio: $\exp(-0.0192)$ 0.981

So, the odds of death decrease by ~1.9% for every 1% increase in the Hispanic population.

percent_insufficient_sleep: For every 1% increase in the rate of insufficient sleep (adults who report not getting enough sleep), the log-odds of death increase by 0.0444.

Converting to odds ratio: $\exp(0.0444)$ 1.0454

So, the odds of death increase by about 4.54% for every 1% increase in the rate of insufficient sleep.

median_household_income: For each \$1 increase in median household income, the log-odds of death decrease by -1.084392e-05.

Convert to a meaningful unit like \$10,000: $10000 * -1.084392e-05 = -0.1084$

Converting to odds ratio: $\exp(-0.1084)$ 0.897

So, the odds of COVID-19 death decrease by about 10.3% for every \$10,000 increase in median household income.

0.1.8 Q1.5 (10pts)

Apply analysis of deviance to (1) evaluate the goodness of fit of the model and (2) compare the model to the intercept-only model. **Solution:**

```

null_model <- glm(cbind(deaths, confirmed - deaths) ~ 1,
                  family = binomial(),
                  data = county_data)

anova(null_model, full_model, test = "Chisq")

```

Analysis of Deviance Table

Model 1: cbind(deaths, confirmed - deaths) ~ 1

Model 2: cbind(deaths, confirmed - deaths) ~ state + percent_fair_or_poor_health + percent_smokers + percent_adults_with_obesity + percent_with_access_to_exercise_opportunity + percent_excessive_drinking + percent_uninsured + percent_some_college + percent_unemployed + percent_children_in_poverty + percent_single_parent_households + percent_severe_housing_problems + overcrowding + percent_adults_with_diabetes + percent_food_insecure + percent_insufficient_sleep + percent_uninsured_2 + median_household_income + average_traffic_volume_per_meter_of_major_roadways + percent_homeowners + population_2 + percent_less_than_18_years_of_age + percent_65_and_over + percent_black + percent_asian + percent_hispanic + percent_female + percent_rural

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	1445	3257.7			
2	1368	1842.8	77	1414.9	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

# Extract residual deviance and degrees of freedom
res_dev <- deviance(full_model)
res_df <- df.residual(full_model)

# Perform chi-squared goodness-of-fit test
p_value <- pchisq(res_dev, df = res_df, lower.tail = FALSE)

# Output
cat("Residual deviance:", res_dev, "\n")

```

Residual deviance: 1842.755

```

cat("Residual degrees of freedom:", res_df, "\n")

```

Residual degrees of freedom: 1368


```
cat("Goodness-of-fit p-value:", p_value, "\n")
```

Goodness-of-fit p-value: 1.099787e-16

Model Comparison:

The test returned a p-value of 2.2e-16, indicating that the full model provides a significantly better fit to the data than the null model.

Goodness-of-Fit:

The residual deviance of the full model is 1842.755, and the residual degrees of freedom is 1368. The Goodness-of-fit p-value is extremely small (1.099787e-16), suggesting that the model may not fit the data perfectly .

0.1.9 Q1.6 (10pts)

Perform analysis of deviance to evaluate the significance of each predictor. Display the 10 most significant predictors. **Solution:**

```
library(car)
```

Warning: package 'car' was built under R version 4.3.3

```
# Type II Analysis of Deviance
deviance_table <- Anova(full_model, type = "II", test = "LR")

# Pull out the top 10 by p-value
top_predictors <- as.data.frame(deviance_table) %>%
  rownames_to_column("Predictor") %>%
  arrange(`Pr(>Chisq)`) %>%
  slice(1:10)

print(top_predictors)
```

	Predictor	LR Chisq	Df	Pr(>Chisq)
1	state	571.076667	50	1.491062e-89
2	percent_homeowners	30.984116	1	2.601488e-08
3	percent_insufficient_sleep	19.993884	1	7.769024e-06
4	percent_severe_housing_problems	17.155368	1	3.444346e-05
5	percent_children_in_poverty	15.666979	1	7.553140e-05

6	median_household_income	10.700584	1	1.071017e-03
7	percent_some_college	9.254634	1	2.349003e-03
8	percent_fair_or_poor_health	8.797733	1	3.016051e-03
9	percent_less_than_18_years_of_age	8.750920	1	3.094459e-03
10	percent_smokers	6.022785	1	1.412235e-02

0.1.10 Q1.7 (5pts)

Construct confidence intervals of regression coefficients. **Solution:**

```
confint(full_model)
```

	2.5 %	97.5 %
(Intercept)	-1.540685e+00	4.829928e+00
stateAlaska	-3.753299e+00	7.665854e-01
stateArizona	-1.121919e-01	1.326975e+00
stateArkansas	-2.272222e+00	-2.941005e-01
stateCalifornia	-4.270811e-01	8.432042e-01
stateColorado	-8.047425e-02	9.558273e-01
stateConnecticut	-4.431493e-01	5.408817e-01
stateDelaware	-1.343063e+00	6.491602e-02
stateDistrict of Columbia	-2.385522e+00	-3.452739e-01
stateFlorida	-2.045630e-01	7.356315e-01
stateGeorgia	-1.575974e-01	6.138168e-01
stateHawaii	-3.239788e+00	-4.080654e-01
stateIdaho	-1.491142e+00	1.890745e-01
stateIllinois	-2.839137e-01	5.952862e-01
stateIndiana	-2.054079e-02	8.447684e-01
stateIowa	-1.301477e+00	2.507241e-01
stateKansas	-3.042060e-01	1.020971e+00
stateKentucky	-1.643561e-01	8.894116e-01
stateLouisiana	-2.837727e-01	5.505662e-01
stateMaine	-1.538751e+00	2.289083e-01
stateMaryland	-1.256754e+00	-2.257937e-01
stateMassachusetts	-1.418191e+00	-3.792222e-01
stateMichigan	-1.674387e-01	6.492574e-01
stateMinnesota	-9.984654e-01	3.106177e-01
stateMississippi	-2.569089e-01	7.032953e-01
stateMissouri	-8.618337e-01	1.558494e-01
stateMontana	-1.068580e+00	7.935561e-01
stateNebraska	-1.524163e+00	7.176783e-01

stateNevada	-2.755078e-01	8.345940e-01
stateNew Hampshire	-3.354740e+00	-8.509625e-01
stateNew Jersey	-4.091492e-01	4.592997e-01
stateNew Mexico	-4.112221e-01	1.809014e+00
stateNew York	-1.403926e+00	-4.363713e-01
stateNorth Carolina	-1.123934e+00	-1.511085e-01
stateNorth Dakota	-2.782881e+00	3.734572e-01
stateOhio	-5.211860e-01	3.949778e-01
stateOklahoma	2.753046e-01	1.419962e+00
stateOregon	-5.767115e-01	6.730255e-01
statePennsylvania	-1.252486e+00	-3.499987e-01
stateRhode Island	-3.614905e+00	-1.077287e+00
stateSouth Carolina	-1.012535e+00	-3.916322e-02
stateSouth Dakota	-2.568364e+00	5.247816e-01
stateTennessee	-9.945980e-01	-1.098143e-01
stateTexas	9.571634e-02	1.595777e+00
stateUtah	-1.982670e+00	-8.921095e-02
stateVermont	-2.328611e+00	-2.914186e-01
stateVirginia	-2.181136e+00	-9.283012e-01
stateWashington	7.105650e-02	1.028006e+00
stateWest Virginia	-3.353514e+00	-3.202647e-01
stateWisconsin	-7.521121e-01	3.500790e-01
stateWyoming	-4.065085e+02	-4.687629e+02
percent_fair_or_poor_health	-9.034904e-02	-1.839060e-02
percent_smokers	-7.751397e-02	-8.679085e-03
percent_adults_with_obesity	-2.246740e-02	1.113176e-02
percent_with_access_to_exercise_opportunities	-5.745291e-03	5.231292e-03
percent_excessive_drinking	-4.111152e-02	1.042741e-02
percent_uninsured	-3.573039e-01	-1.480706e-02
percent_some_college	-2.689944e-02	-5.803777e-03
percent_unemployed	-3.626838e-02	1.451401e-01
percent_children_in_poverty	1.526815e-02	4.528901e-02
percent_single_parent_households	-2.741900e-02	-2.963883e-03
percent_severe_housing_problems	-6.578986e-02	-2.361278e-02
overcrowding	-3.071034e-02	7.016017e-02
percent_adults_with_diabetes	-4.908876e-02	5.180565e-03
percent_food_insecure	-1.891933e-01	-2.105405e-03
percent_insufficient_sleep	2.490233e-02	6.388899e-02
percent_uninsured_2	-1.936819e-02	2.612330e-01
median_household_income	-1.735930e-05	-4.342325e-06
average_traffic_volume_per_meter_of_major_roadways	-7.390708e-05	1.048167e-04
percent_homeowners	-4.605639e-02	-2.196463e-02
population_2	-6.804682e-08	-2.338900e-09

percent_less_than_18_years_of_age	-7.990811e-02	-1.626028e-02
percent_65_and_over	-1.360161e-02	3.253736e-02
percent_black	-2.712684e-03	2.024150e-02
percent_asian	-1.233780e-02	1.112980e-02
percent_hispanic	-3.569677e-02	-2.750001e-03
percent_female	-4.049216e-03	1.044219e-01
percent_rural	-2.958610e-03	5.070711e-03

```
# ALTERNATIVE SOLUTION (commented out)
# library(broom)
# library(tidyverse)

# Profile-likelihood CIs throws out warnings
# ci_profile <- confint(full_model) %>%
#   as_tibble(rownames = "term") %>%
#   rename(profile_low = `2.5 %`, profile_high = `97.5 %`)

# Wald CIs (faster approximation)
# ci_wald <- tidy(full_model, conf.int = TRUE, conf.level = 0.95) %>%
#   select(term, estimate, std.error,
#          wald_low = conf.low, wald_high = conf.high,
#          p.value)

# Combine results
# full_results <- ci_wald %>%
#   left_join(ci_profile, by = "term") %>%
#   arrange(p.value)

# Display top predictors
# full_results %>%
#   head(10) %>%
#   knitr::kable(digits = 4)
```

0.1.11 Q1.8 (5pts)

Plot the deviance residuals against the fitted values. Are there potential outliers? **Solution:**

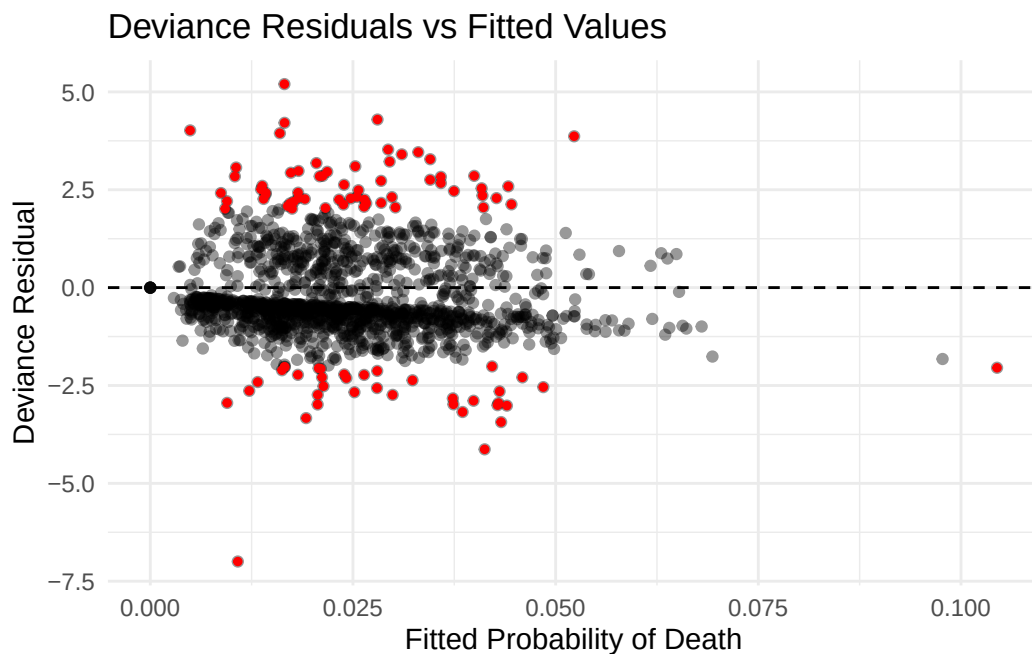
```
library(ggplot2)

# Extract deviance residuals and fitted probabilities
resid_dev <- residuals(full_model, type = "deviance")
```

```
fitted_prob <- fitted(full_model)

# Build a data frame
plot_df <- data.frame(
  fitted = fitted_prob,
  dev_resid = resid_dev
)

# 3. Plot
ggplot(plot_df, aes(x = fitted, y = dev_resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  # highlight points with |resid| > 2
  geom_point(data = subset(plot_df, abs(dev_resid) > 2),
    color = "red", size = 1) +
  labs(
    title = "Deviance Residuals vs Fitted Values",
    x = "Fitted Probability of Death",
    y = "Deviance Residual"
  ) +
  theme_minimal()
```



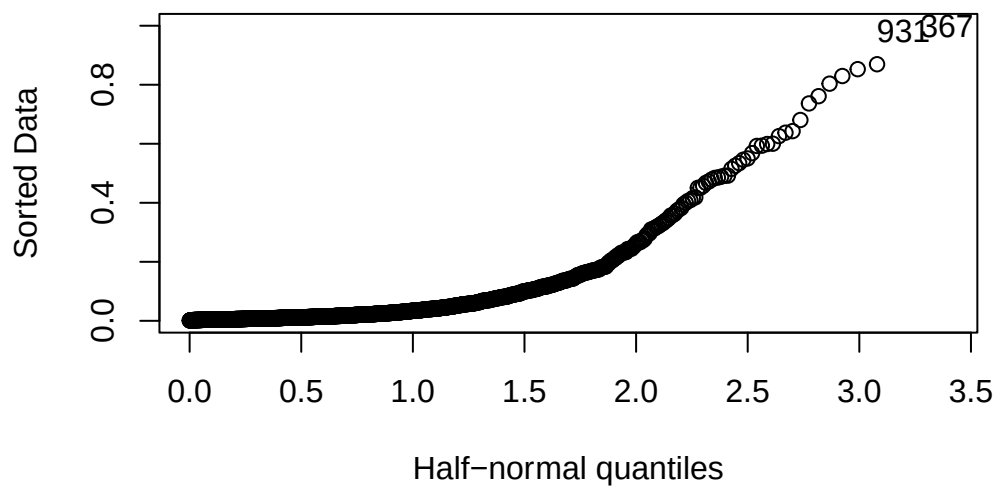
0.1.12 Q1.9 (5pts)

Plot the half-normal plot. Are there potential outliers in predictor space? **Solution:**

```
library(faraway)
```

Warning: package 'faraway' was built under R version 4.3.3

```
halfnorm(hatvalues(full_model))
```



Based on the plot, the points 931 and 367 are potential outliers in predictor space.

```
# ALTERNATIVE SOLUTION (commented out)
# Compute leverage (hat) values
# lev <- hatvalues(full_model)
# n <- length(lev)
# lev_ord <- sort(lev)

# hnq <- qnorm( (2*seq_len(n) - 1) / (2*n) ) # half-normal
# hnq <- sort(abs(hnq))
```

```

# df <- data.frame(hnq, lev = lev_ord)

#ggplot(df, aes(hnq, lev)) +
#  geom_point(alpha = .6) +
#  geom_abline(intercept = 0,
#              slope = sum(lev_ord*hnq)/sum(hnq^2),
#              linetype = "dashed") +
#  labs(x = "Half-normal quantiles",
#       y = "Leverage (hat-values)",
#       title = "Half-Normal plot of leverage")

# Identify high-leverage points
# p <- length(coef(full_model))
# n <- nobs(full_model)
# thresh <- 2 * (p + 1) / n

# lev <- hatvalues(full_model)
# high_lev_idx <- which(lev > thresh)

# View the top offenders
# data.frame(
#   county = county_data$county[high_lev_idx],
#   state = county_data$state[high_lev_idx],
#   leverage = lev[high_lev_idx]
# ) %>%
#   arrange(desc(leverage)) %>%
#   head(10)

```

0.1.13 Q1.10 (10pts)

Find the best sub-model using the AIC criterion. **Solution:**

```

library(MASS)

# Run stepwise AIC (both directions)
step_model <- stepAIC(
  object = full_model,
  direction = "both",
  trace = FALSE
)

```

```
# Show the final selected formula
cat("Selected model formula:\n")
```

Selected model formula:

```
print(formula(step_model))
```

```
cbind(deaths, confirmed - deaths) ~ state + percent_fair_or_poor_health +
  percent_smokers + percent_excessive_drinking + percent_uninsured +
  percent_some_college + percent_unemployed + percent_children_in_poverty +
  percent_single_parent_households + percent_severe_housing_problems +
  percent_adults_with_diabetes + percent_food_insecure + percent_insufficient_sleep +
  percent_uninsured_2 + median_household_income + percent_homeowners +
  population_2 + percent_less_than_18_years_of_age + percent_black +
  percent_hispanic + percent_female
```

```
# Report its AIC
cat("\nAIC of selected model:", AIC(step_model), "\n")
```

AIC of selected model: 3870.062

```
# Inspect coefficient estimates
summary(step_model)
```

Call:

```
glm(formula = cbind(deaths, confirmed - deaths) ~ state + percent_fair_or_poor_health +
  percent_smokers + percent_excessive_drinking + percent_uninsured +
  percent_some_college + percent_unemployed + percent_children_in_poverty +
  percent_single_parent_households + percent_severe_housing_problems +
  percent_adults_with_diabetes + percent_food_insecure + percent_insufficient_sleep +
  percent_uninsured_2 + median_household_income + percent_homeowners +
  population_2 + percent_less_than_18_years_of_age + percent_black +
  percent_hispanic + percent_female, family = binomial(), data = county_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.021e+00	1.433e+00	1.410	0.158473

stateAlaska	-8.993e-01	1.035e+00	-0.869	0.385118	
stateArizona	6.866e-01	3.325e-01	2.065	0.038943	*
stateArkansas	-1.137e+00	4.897e-01	-2.321	0.020263	*
stateCalifornia	2.819e-01	2.924e-01	0.964	0.335047	
stateColorado	4.699e-01	2.520e-01	1.865	0.062219	.
stateConnecticut	3.094e-02	2.468e-01	0.125	0.900234	
stateDelaware	-6.128e-01	3.528e-01	-1.737	0.082401	.
stateDistrict of Columbia	-1.449e+00	4.876e-01	-2.972	0.002957	**
stateFlorida	3.012e-01	2.174e-01	1.386	0.165836	
stateGeorgia	1.965e-01	1.950e-01	1.008	0.313524	
stateHawaii	-1.496e+00	6.594e-01	-2.269	0.023270	*
stateIdaho	-5.607e-01	4.147e-01	-1.352	0.176345	
stateIllinois	1.146e-01	2.090e-01	0.548	0.583525	
stateIndiana	4.209e-01	2.138e-01	1.969	0.048969	*
stateIowa	-4.222e-01	3.728e-01	-1.133	0.257375	
stateKansas	3.870e-01	3.225e-01	1.200	0.230107	
stateKentucky	3.834e-01	2.605e-01	1.472	0.141122	
stateLouisiana	9.962e-02	2.042e-01	0.488	0.625716	
stateMaine	-5.414e-01	4.408e-01	-1.228	0.219328	
stateMaryland	-7.691e-01	2.587e-01	-2.973	0.002947	**
stateMassachusetts	-9.356e-01	2.562e-01	-3.651	0.000261	***
stateMichigan	2.461e-01	1.974e-01	1.247	0.212472	
stateMinnesota	-2.857e-01	3.267e-01	-0.875	0.381707	
stateMississippi	2.407e-01	2.428e-01	0.991	0.321486	
stateMissouri	-3.261e-01	2.513e-01	-1.298	0.194302	
stateMontana	-2.997e-02	4.637e-01	-0.065	0.948464	
stateNebraska	-2.313e-01	5.469e-01	-0.423	0.672328	
stateNevada	3.064e-01	2.595e-01	1.181	0.237790	
stateNew Hampshire	-1.905e+00	6.151e-01	-3.097	0.001955	**
stateNew Jersey	7.690e-03	2.114e-01	0.036	0.970988	
stateNew Mexico	8.546e-01	5.283e-01	1.618	0.105757	
stateNew York	-8.936e-01	2.402e-01	-3.721	0.000198	***
stateNorth Carolina	-6.336e-01	2.456e-01	-2.580	0.009889	**
stateNorth Dakota	-9.406e-01	7.626e-01	-1.234	0.217385	
stateOhio	-6.367e-02	2.204e-01	-0.289	0.772649	
stateOklahoma	8.593e-01	2.756e-01	3.118	0.001821	**
stateOregon	4.885e-02	3.077e-01	0.159	0.873865	
statePennsylvania	-8.145e-01	2.211e-01	-3.685	0.000229	***
stateRhode Island	-2.206e+00	6.231e-01	-3.540	0.000400	***
stateSouth Carolina	-5.282e-01	2.452e-01	-2.154	0.031207	*
stateSouth Dakota	-6.964e-01	7.453e-01	-0.934	0.350068	
stateTennessee	-5.583e-01	2.243e-01	-2.489	0.012802	*
stateTexas	8.853e-01	3.478e-01	2.545	0.010914	*

stateUtah	-9.414e-01	4.596e-01	-2.048	0.040539	*
stateVermont	-1.165e+00	5.069e-01	-2.298	0.021548	*
stateVirginia	-1.544e+00	3.166e-01	-4.878	1.07e-06	***
stateWashington	5.204e-01	2.322e-01	2.241	0.025016	*
stateWest Virginia	-1.539e+00	7.336e-01	-2.098	0.035945	*
stateWisconsin	-1.945e-01	2.676e-01	-0.727	0.467488	
stateWyoming	-1.369e+01	3.455e+02	-0.040	0.968398	
percent_fair_or_poor_health	-4.910e-02	1.756e-02	-2.796	0.005181	**
percent_smokers	-4.983e-02	1.676e-02	-2.972	0.002956	**
percent_excessive_drinking	-1.623e-02	1.134e-02	-1.431	0.152413	
percent_uninsured	-1.770e-01	8.567e-02	-2.066	0.038825	*
percent_some_college	-1.867e-02	4.810e-03	-3.882	0.000104	***
percent_unemployed	7.914e-02	4.122e-02	1.920	0.054887	.
percent_children_in_poverty	3.338e-02	6.727e-03	4.962	6.96e-07	***
percent_single_parent_households	-1.713e-02	6.018e-03	-2.846	0.004429	**
percent_severe_housing_problems	-4.184e-02	8.959e-03	-4.670	3.02e-06	***
percent_adults_with_diabetes	-2.365e-02	1.304e-02	-1.814	0.069649	.
percent_food_insecure	-1.071e-01	4.229e-02	-2.533	0.011300	*
percent_insufficient_sleep	4.044e-02	9.443e-03	4.283	1.85e-05	***
percent_uninsured_2	1.170e-01	7.025e-02	1.665	0.095859	.
median_household_income	-1.056e-05	2.699e-06	-3.914	9.07e-05	***
percent_homeowners	-3.626e-02	4.992e-03	-7.263	3.78e-13	***
population_2	-3.139e-08	1.509e-08	-2.080	0.037558	*
percent_less_than_18_years_of_age	-5.939e-02	1.098e-02	-5.408	6.39e-08	***
percent_black	8.630e-03	5.464e-03	1.580	0.114212	
percent_hispanic	-2.193e-02	7.726e-03	-2.838	0.004542	**
percent_female	5.759e-02	2.234e-02	2.578	0.009950	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 3257.7 on 1445 degrees of freedom
 Residual deviance: 1846.2 on 1375 degrees of freedom
 AIC: 3870.1

Number of Fisher Scoring iterations: 14

Best AIC model: cbind(deaths, confirmed - deaths) ~ state + percent_fair_or_poor_health + percent_smokers + percent_excessive_drinking + percent_uninsured + percent_some_college + percent_unemployed + percent_children_in_poverty + percent_single_parent_households + percent_severe_housing_problems + percent_adults_with_diabetes + percent_food_insecure

+ percent_insufficient_sleep + percent_uninsured_2 + median_household_income + percent_homeowners + population_2 + percent_less_than_18_years_of_age + percent_black + percent_hispanic + percent_female

Final AIC = 3870.06, a drop of ~10.6 units from the full model.

0.1.14 Q1.11 (15pts)

Find the best sub-model using the lasso with cross validation. **Solution:**

```
library(glmnet)
```

Warning: package 'Matrix' was built under R version 4.3.1

```
# Build the design matrix and response
x <- model.matrix(
  deaths/confirmed ~ state
  + percent_fair_or_poor_health
  + percent_smokers
  + percent_adults_with_obesity
  + percent_with_access_to_exercise_opportunities
  + percent_excessive_drinking
  + percent_uninsured
  + percent_some_college
  + percent_unemployed
  + percent_children_in_poverty
  + percent_single_parent_households
  + percent_severe_housing_problems
  + overcrowding
  + percent_adults_with_diabetes
  + percent_food_insecure
  + percent_insufficient_sleep
  + percent_uninsured_2
  + median_household_income
  + average_traffic_volume_per_meter_of_major_roadways
  + percent_homeowners
  + population_2
  + percent_less_than_18_years_of_age
  + percent_65_and_over
  + percent_black
```

```

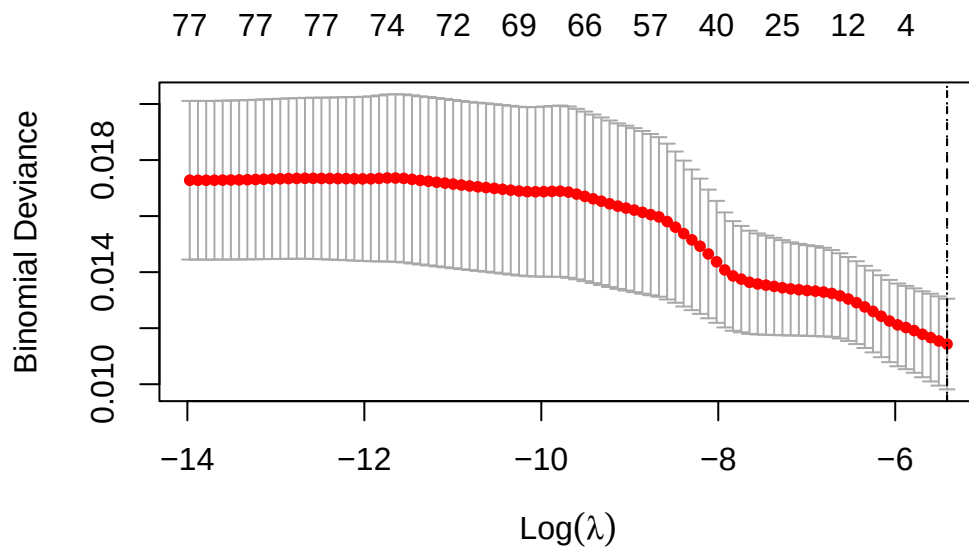
+ percent_asian
+ percent_hispanic
+ percent_female
+ percent_rural,
data = county_data
)[, -1]

## grouped-binomial response
y <- cbind(
  deaths      = county_data$deaths,
  survivors   = county_data$confirmed - county_data$deaths
)

## 10-fold CV lasso
set.seed(123)
cv_lasso <- cv.glmnet(
  x, y,
  family      = "binomial",
  type.measure = "deviance"
)

## visualise & extract coefficients
plot(cv_lasso)
abline(v = log(c(cv_lasso$lambda.min,
                  cv_lasso$lambda.1se)),
       lty = 2)

```



```
best_coefs <- coef(cv_lasso, s = "lambda.min")
nonzero     <- which(best_coefs != 0)

lasso_results <- data.frame(
  term      = rownames(best_coefs)[nonzero],
  estimate  = as.numeric(best_coefs[nonzero])
)
print(lasso_results, row.names = FALSE)
```

```
      term estimate
(Intercept) 3.730008
```

Since CV lasso selects a model with just the intercept, we can check the coefficients at the next 5 smaller lambdas (more predictors).

```
# index of lambda.min on the path
k_min <- match(cv_lasso$lambda.min, cv_lasso$glmnet.fit$lambda)

# pick the next 5 smaller lambdas (more predictors)
k_seq <- k_min + 0:5
k_seq <- k_seq[k_seq <= length(cv_lasso$glmnet.fit$lambda)]
```

```

lambda_seq <- cv_lasso$glmnet.fit$lambda[k_seq]

results <- lapply(lambda_seq, function(lam) {
  beta <- coef(cv_lasso$glmnet.fit, s = lam)
  nz <- which(beta != 0)
  data.frame(
    lambda      = lam,
    n_nonzero    = length(nz),
    cv_deviance = cv_lasso$cvm[match(lam, cv_lasso$lambda)],
    term        = rownames(beta)[nz],
    estimate     = as.numeric(beta[nz]),
    row.names    = NULL
  )
})

results_df <- do.call(rbind, results)

# show one table per lambda
library(dplyr)
results_df %>%
  group_by(lambda, n_nonzero, cv_deviance) %>%
  summarise(terms = paste(term, collapse = ", "),
    .groups = "drop") %>%
  arrange(lambda) %>%
  knitr::kable(digits = 4)

```

lambda	n_nonzero	cv_deviance	terms
0.0028	5	0.0120	(Intercept), stateMichigan, stateWashington, percent_food_insecure, percent_homeowners
0.0031	4	0.0119	(Intercept), stateWashington, percent_food_insecure, percent_homeowners
0.0034	3	0.0118	(Intercept), stateWashington, percent_food_insecure
0.0037	2	0.0117	(Intercept), percent_food_insecure
0.0041	2	0.0115	(Intercept), percent_food_insecure
0.0045	1	0.0114	(Intercept)

0.2 Q2. Odds ratios (20pts)

Consider a 2×2 contingency table from a prospective study in which people who were or were not exposed to some pollutant are followed up and, after several years, categorized according

to the presense or absence of a disease. Following table shows the probabilities for each cell. The odds of disease for either exposure group is $O_i = \pi_i/(1 - \pi_i)$, for $i = 1, 2$, and so the odds ratio is

$$\phi = \frac{O_1}{O_2} = \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)}$$

is a measure of the relative likelihood of disease for the exposed and not exposed groups.

	Diseased	Not diseased
Exposed	π_1	$1 - \pi_1$
Not exposed	π_2	$1 - \pi_2$

0.2.1 Q2.1 (10pts)

For the simple logistic model

$$\pi_i = \frac{e^{\beta_i}}{1 + e^{\beta_i}},$$

show that if there is no difference between the exposed and not exposed groups (i.e., $\beta_1 = \beta_2$), then $\phi = 1$.

Solution:

The odds for group i are

$$\begin{aligned} O_i &= \frac{\pi_i}{1 - \pi_i} \\ &= \frac{\frac{e^{\beta_i}}{1 + e^{\beta_i}}}{1 - \frac{e^{\beta_i}}{1 + e^{\beta_i}}} \\ &= \frac{e^{\beta_i}}{1 + e^{\beta_i}} \cdot \frac{1 + e^{\beta_i}}{1} = e^{\beta_i}. \end{aligned}$$

So the odds-ratio is

$$\phi = \frac{O_1}{O_2} = \frac{e^{\beta_1}}{e^{\beta_2}} = e^{\beta_1 - \beta_2}.$$

If $\beta_1 = \beta_2$, then

$$\phi = e^{\beta_1 - \beta_1} = e^0 = 1$$

0.2.2 Q2.2(10pts)

Consider J 2×2 tables, one for each level x_j of a factor, such as age group, with $j = 1, \dots, J$. For the logistic model

$$\pi_{ij} = \frac{e^{\alpha_i + \beta_i x_j}}{1 + e^{\alpha_i + \beta_i x_j}}, \quad i = 1, 2, \quad j = 1, \dots, J.$$

Show that $\log \phi$ is constant over all tables if $\beta_1 = \beta_2$.

Solution:

Given

$$\pi_{ij} = \frac{e^{\alpha_i + \beta_i x_j}}{1 + e^{\alpha_i + \beta_i x_j}}$$

the odds for group (i) at level (x_j) are

$$O_{ij} = \frac{\pi_{ij}}{1 - \pi_{ij}} = e^{\alpha_i + \beta_i x_j}.$$

The odds-ratio for each (j) is

$$\phi_j = \frac{O_{1j}}{O_{2j}} = \frac{e^{\alpha_1 + \beta_1 x_j}}{e^{\alpha_2 + \beta_2 x_j}} = e^{(\alpha_1 - \alpha_2) + (\beta_1 - \beta_2) x_j}.$$

So the log-odds-ratio is

$$\log \phi_j = (\alpha_1 - \alpha_2) + (\beta_1 - \beta_2) x_j.$$

If ($\beta_1 = \beta_2$), then

$$\log \phi_j = (\alpha_1 - \alpha_2),$$

which no longer depends on (x_j). We can conclude the log-odds-ratio is **constant** across all (j).

0.3 Q3. ELMR Chapter 4 Exercise 3 (30pts)

The infert dataset presents data from a study of secondary infertility (failure to conceive after at least one previous conception). The factors of interest are induced abortions and spontaneous abortions (e.g., miscarriages). The study matched each case of infertility with two controls who were not infertile, matching on age, education and parity (number of prior pregnancies).

0.3.1 Q3(a) Construct cross-classified tables by number of spontaneous and induced abortions separately for cases and controls. Comment on the differences between the two tables.

```
library(tidyverse)
library(broom)
library(survival)

data(infert, package = "datasets")
glimpse(infert)
```

Rows: 248

Columns: 8

\$ education	<fct>	0-5yrs, 0-5yrs, 0-5yrs, 0-5yrs, 6-11yrs, 6-11yrs, 6-11yr~
\$ age	<dbl>	26, 42, 39, 34, 35, 36, 23, 32, 21, 28, 29, 37, 31, 29, ~
\$ parity	<dbl>	6, 1, 6, 4, 3, 4, 1, 2, 1, 2, 2, 4, 1, 3, 2, 2, 5, 1, 3~
\$ induced	<dbl>	1, 1, 2, 2, 1, 2, 0, 0, 0, 0, 1, 2, 1, 2, 1, 2, 2, 0, 2~
\$ case	<dbl>	1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
\$ spontaneous	<dbl>	2, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 1, 1~
\$ stratum	<int>	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
\$ pooled.stratum	<dbl>	3, 1, 4, 2, 32, 36, 6, 22, 5, 19, 20, 37, 9, 29, 21, 18~

```
# Nicer labels
infert <- infert %>%
  mutate(case_lbl = if_else(case == 1, "Case (infertile)", "Control"))

# Cross-tab for spontaneous abortions
table_spont <- with(infert, table(case_lbl, spontaneous))
table_induc <- with(infert, table(case_lbl, induced))

table_spont
```

	spontaneous		
case_lbl	0	1	2
Case (infertile)	28	31	24
Control	113	40	12

```
table_induc
```

	induced		
case_lbl	0	1	2
Case (infertile)	47	23	13
Control	96	45	24

```
prop.table(table_spont, margin = 1)
```

	spontaneous		
case_lbl	0	1	2
Case (infertile)	0.33734940	0.37349398	0.28915663
Control	0.68484848	0.24242424	0.07272727

```
prop.table(table_induc, margin = 1)
```

	induced		
case_lbl	0	1	2
Case (infertile)	0.5662651	0.2771084	0.1566265
Control	0.5818182	0.2727273	0.1454545

Spontaneous abortions:

Among infertile cases the percentages are roughly split with 0 abortions 34 %, 1 37 %, 2 29 %.

Among controls most women 68 % report zero spontaneous abortions, and only 7 % report two. a history of more spontaneous abortions is noticeably more common in cases than controls, suggesting a possible association with infertility.

Induced abortions:

The share with zero induced abortions is similar in both groups (~57 %), but the case group shows slightly higher proportions at 1 and 2 induced abortions.

Induced abortions may still matter, but the contrast is weaker than for spontaneous abortions.

0.3.2 Q3(b) Fit a binary response model with only spontaneous and induced as predictors.

Determine the statistical significance of these predictors. Express the effects of the predictors in terms of odds.

```
mod_b <- glm(case ~ spontaneous + induced, family = binomial, data = infert)
tidy(mod_b, exponentiate = TRUE, conf.int = TRUE)
```

```
# A tibble: 3 x 7
  term          estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    0.181    0.268    -6.38 1.78e-10  0.105    0.300
2 spontaneous    3.31     0.212     5.66 1.54e- 8  2.21     5.09
3 induced        1.52     0.206     2.03 4.20e- 2  1.02     2.28
```

Both variables are significant; the effect of spontaneous abortions is more than twice as strong as that of induced abortions.

0.3.3 Q3(c) Fit a binary response model with only education, age and parity as predictors.

Explain how the significance (or lack thereof) of these predictors should be interpreted.

```
mod_c <- glm(case ~ education + age + parity, family = binomial, data = infert)
tidy(mod_c, exponentiate = TRUE, conf.int = TRUE)
```

```
# A tibble: 5 x 7
  term          estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)    0.428    1.25     -0.680  0.496  0.0359    4.87
2 education6-11yrs 1.05     0.694     0.0664  0.947  0.279    4.50
3 education12+ yrs 1.07     0.718     0.0975  0.922  0.271    4.80
4 age            1.00     0.0272     0.0762  0.939  0.950    1.06
5 parity         1.02     0.117     0.163  0.871  0.807    1.28
```

None of the three predictors are statistically significant. The estimated ORs are close to 1, with very wide CIs that include the null value. This is what we expect: age, education

and parity were the matching variables, so their marginal associations with case status are purposely minimized after matching.

0.3.4 Q3(d) Now put all five predictors in a binary response model. Interpret the results in terms of odds.

```
mod_d <- glm(case ~ spontaneous + induced + education + age + parity,
              family = binomial, data = infert)
tidy(mod_d, exponentiate = TRUE, conf.int = TRUE)
```

```
# A tibble: 7 x 7
  term                estimate std.error statistic  p.value  conf.low  conf.high
  <chr>              <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)        0.317      1.41     -0.814  4.16e- 1  0.0192    5.01
2 spontaneous         7.74      0.310      6.60  4.21e-11  4.33     14.7
3 induced             3.63      0.301      4.28  1.91e- 5  2.05      6.70
4 education6-11yrs    0.352      0.793     -1.32  1.88e- 1  0.0733    1.74
5 education12+ yrs    0.246      0.834     -1.68  9.25e- 2  0.0467    1.30
6 age                 1.04      0.0312     1.27  2.05e- 1  0.979     1.11
7 parity              0.437      0.196     -4.22  2.49e- 5  0.291     0.631
```

Each extra spontaneous abortion increases the odds of infertility by about $7.7 \times$ after adjusting for the other covariates.

Each extra induced abortion triples the odds.

More previous pregnancies reduce the odds. Every additional pregnancy roughly halves the odds.

Adjustment makes the abortion effects even stronger, while parity now clearly shows an inverse relationship with infertility risk.

0.3.5 Q3(e) Fit a matched case control model appropriate to the data. Interpret the output and compare the odds to those found in the previous model.

```
mod_e <- clogit(case ~ spontaneous + induced + education + age + parity +
                 strata(stratum), data = infert)
tidy(mod_e, exponentiate = TRUE, conf.int = TRUE)
```

```
# A tibble: 6 x 7
  term                estimate std.error statistic      p.value conf.low conf.high
  <chr>              <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 spontaneous        7.29      0.352      5.63 0.0000000175      3.65     14.5
2 induced            4.09      0.361      3.91 0.0000938          2.02     8.30
3 education6-11yrs   NA         0         NA    NA          NA         NA
4 education12+ yrs   NA         0         NA    NA          NA         NA
5 age                NA         0         NA    NA          NA         NA
6 parity             NA         0         NA    NA          NA         NA
```

The values and direction are very close to the full ordinary model, which further supports the conclusion that both types of abortions are strongly associated with secondary infertility. Matching variables drop out because their effects are absorbed by the stratum terms.

0.3.6 Q3(f) The spontaneous and induced predictors could be viewed as ordinal due to the grouping in the highest level. Refit the model using ordinal factors rather than numerical variables for these two predictors. Is there evidence that the ordinal representation is necessary?

```
infert_ord <- infert %>%
  mutate(spontaneousF = factor(spontaneous, ordered = TRUE),
         inducedF      = factor(induced,      ordered = TRUE))

mod_f <- glm(case ~ spontaneousF + inducedF + education + age + parity,
             family = binomial, data = infert_ord)
tidy(mod_f)
```

```
# A tibble: 9 x 5
  term                estimate std.error statistic  p.value
  <chr>              <dbl>    <dbl>    <dbl>    <dbl>
1 (Intercept)        2.20      1.50      1.46 1.44e- 1
2 spontaneousF.L      2.90      0.454      6.38 1.80e-10
3 spontaneousF.Q      0.0150    0.282      0.0533 9.58e- 1
4 inducedF.L          1.83      0.437      4.20 2.70e- 5
5 inducedF.Q          0.0302    0.291      0.104 9.17e- 1
6 education6-11yrs   -1.04      0.793     -1.31 1.90e- 1
7 education12+ yrs   -1.40      0.835     -1.67 9.41e- 2
8 age                0.0393    0.0316      1.24 2.13e- 1
9 parity            -0.830    0.198     -4.20 2.70e- 5
```

```
# Compare using a likelihood-ratio test
anova(mod_d, mod_f, test = "LRT")
```

Analysis of Deviance Table

```
Model 1: case ~ spontaneous + induced + education + age + parity
Model 2: case ~ spontaneousF + inducedF + education + age + parity
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      241      257.80
2      239      257.79  2  0.012869  0.9936
```

```
# Compare AIC
AIC(mod_d, mod_f)
```

	df	AIC
mod_d	7	271.7977
mod_f	9	275.7848

A likelihood-ratio test comparing the ordinal-factor model to the simpler numeric-score model yields Deviance = 0.013 on 2 df with p-value = 0.9936, and the AIC is slightly worse (275.8 vs 271.8). Therefore, there is no evidence that the extra flexibility of treating the variables as ordered factors improves fit, so the simple linear representation is the better option.